

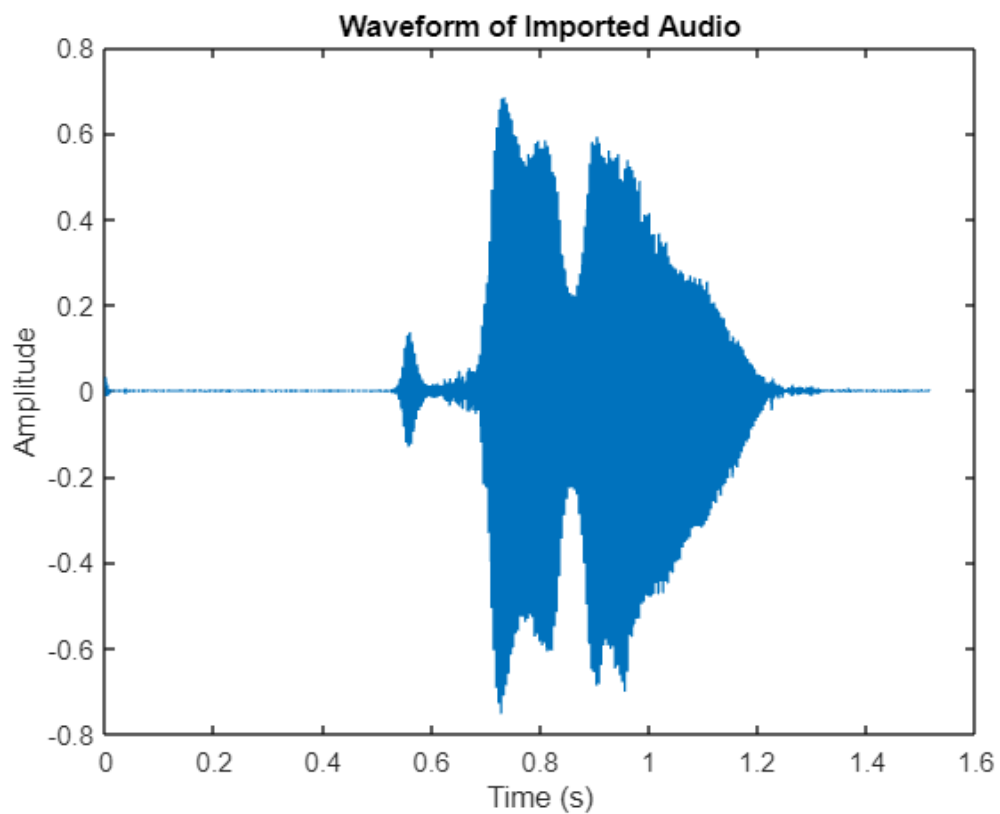
Load the Signal

```
[signal,fs]=audioread("s3.wav");  
signal, fs
```

```
signal = 18944×1  
    0.0312  
    0.0210  
    0.0117  
    0.0176  
    0.0122  
    0.0059  
    0.0088  
    0.0112  
    0.0049  
    0.0054  
        ⋮  
fs = 12500
```

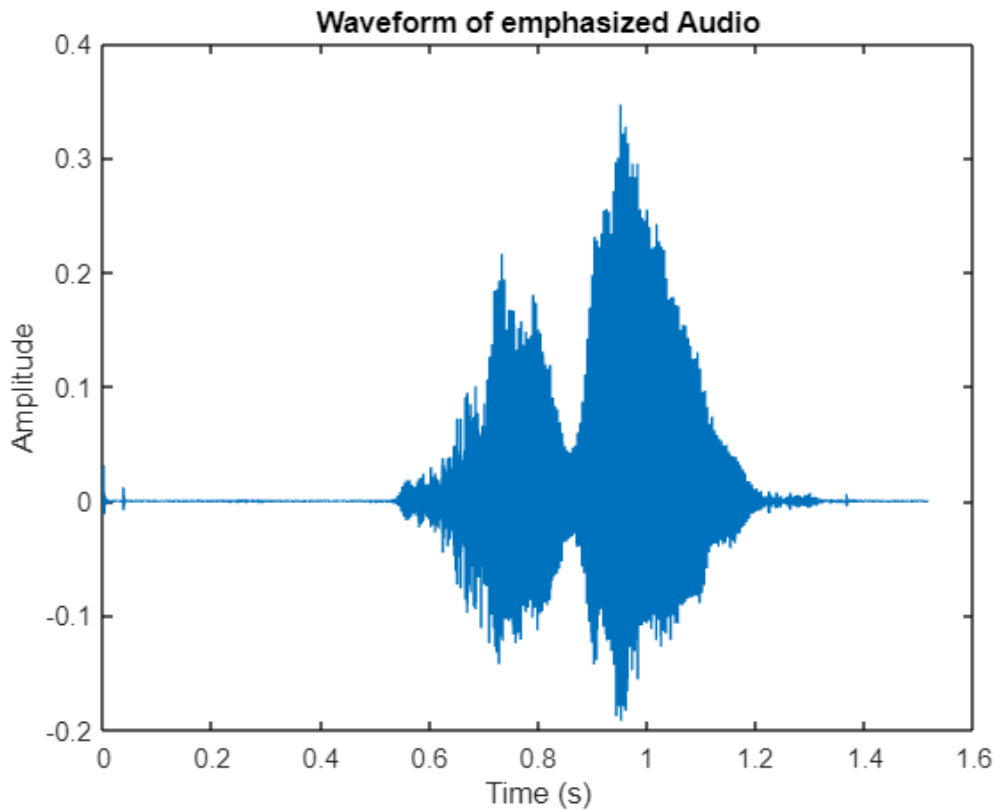
Visualizing the Signal

```
t = (0:length(signal)-1)/fs; % Time axis  
plot(t, signal);  
xlabel('Time (s)');  
ylabel('Amplitude');  
title('Waveform of Imported Audio');
```



Preemphasising

```
alpha = 0.97;
emphasized = preemphasis(signal, alpha);
plot(t, emphasized);
xlabel('Time (s)');
ylabel('Amplitude');
title('Waveform of emphasized Audio');
```



Framing

```
frame_length = 256;
hop_size=100;
frames = frame_signal(emphasized, frame_length, hop_size);
frames
```

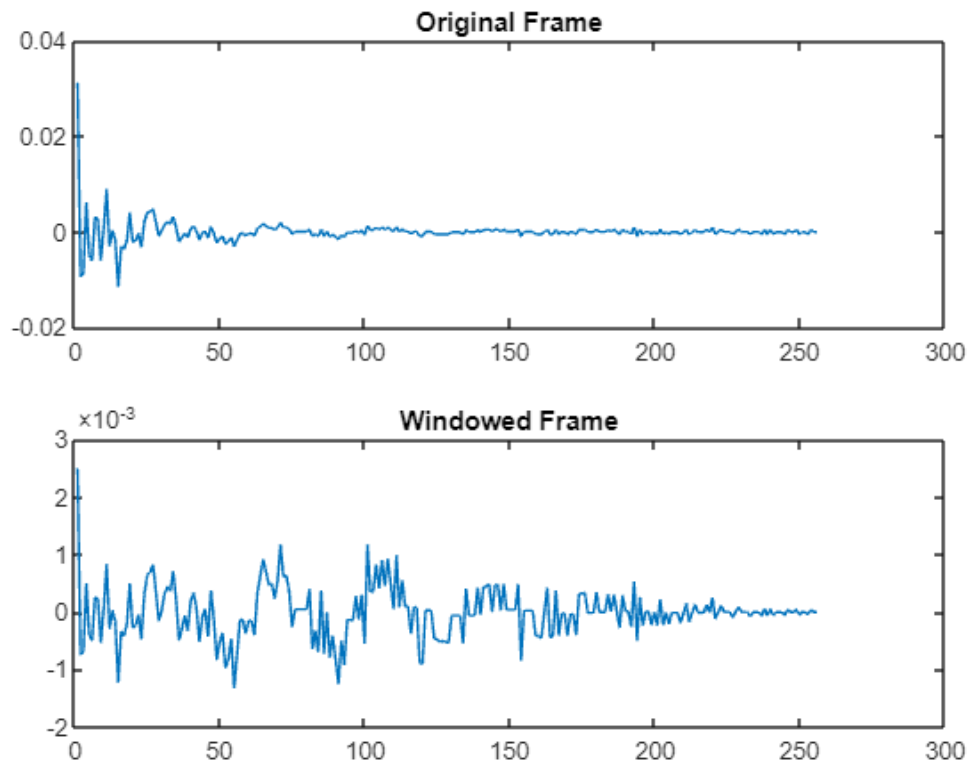
```
frames = 256x187
    0.0312    0.0013   -0.0005         0   -0.0005   -0.0005         0    0.0005 ...
   -0.0093    0.0004    0.0005         0    0.0005   -0.0010   -0.0005   -0.0005
   -0.0086    0.0004   -0.0005    0.0005         0    0.0005    0.0005    0.0005
    0.0062    0.0009   -0.0000   -0.0005   -0.0005         0         0         0
   -0.0048    0.0004   -0.0005    0.0010    0.0005         0         0   -0.0005
   -0.0060    0.0010   -0.0000   -0.0005         0   -0.0005    0.0005    0.0005
    0.0031    0.0005   -0.0000   -0.0005         0   -0.0000   -0.0005    0.0005
    0.0027    0.0010   -0.0000    0.0005         0   -0.0000         0   -0.0005
   -0.0060    0.0005   -0.0005   -0.0005   -0.0005   -0.0000         0         0
    0.0006    0.0001   -0.0000         0    0.0005    0.0005         0   -0.0005
```

⋮

Applying Hamming Window

```
windowed_frames=apply_window(frames);
```

```
frameIdx = 1;  
subplot(2,1,1);  
plot(frames(:, frameIdx));  
title('Original Frame');  
  
subplot(2,1,2);  
plot(windowed_frames(:, frameIdx));  
title('Windowed Frame');
```



```
[frame_length_windowed,number_of_frames]=size(windowed_frames);  
frame_length_windowed
```

```
frame_length_windowed = 256
```

```
number_of_frames
```

```
number_of_frames = 187
```

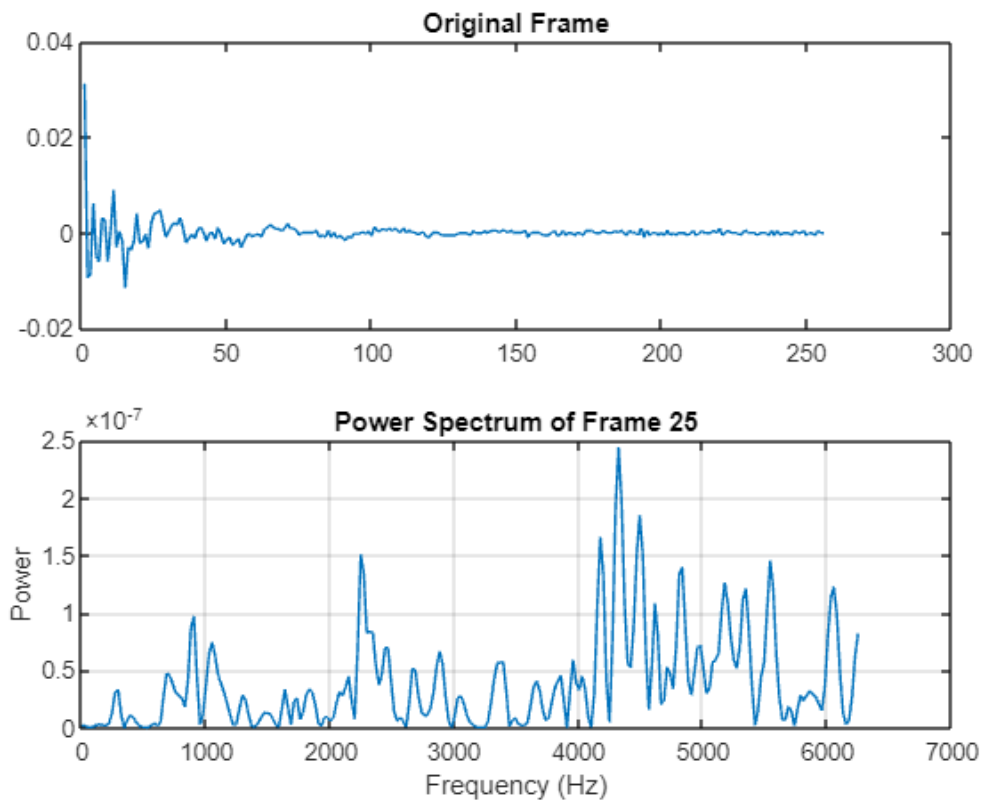
Compute the Power Spectrum

```
nfft = 512;
power_spectrum = compute_power_spectrum(windowed_frames, nfft)
```

```
power_spectrum = 257x187
0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000 ...
0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
...
```

```
frameIdx = 25;
freq_axis = (0:nfft/2) * fs / nfft;

plot(freq_axis, power_spectrum(:, frameIdx));
xlabel('Frequency (Hz)');
ylabel('Power');
title(sprintf('Power Spectrum of Frame %d', frameIdx));
grid on;
```

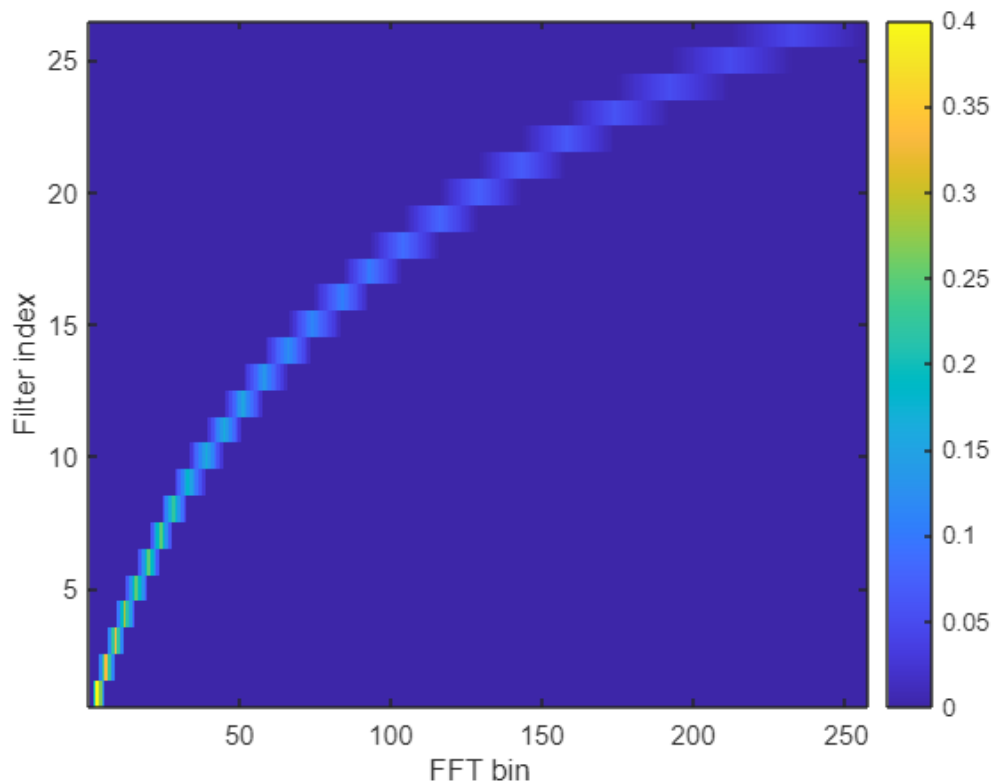


Creating Mel Filter Bank and Applying

```
Num_of_Mel_filters=26;  
Melbank_filter=melfb(Num_of_Mel_filters,nfft,fs)
```

```
Melbank_filter = 26×257  
    0    0.2000    0.4000    0.2667    0.1333    0    0    0 ...  
    0    0    0    0.1111    0.2222    0.3333    0.2222    0.1111  
    0    0    0    0    0    0    0.1111    0.2222  
    0    0    0    0    0    0    0    0  
    0    0    0    0    0    0    0    0  
    0    0    0    0    0    0    0    0  
    0    0    0    0    0    0    0    0  
    0    0    0    0    0    0    0    0  
    0    0    0    0    0    0    0    0  
    ⋮
```

```
figure;  
imagesc(Melbank_filter);  
axis xy;  
xlabel("FFT bin")  
ylabel("Filter index")  
colorbar;
```



```
figure;  
hold on;
```

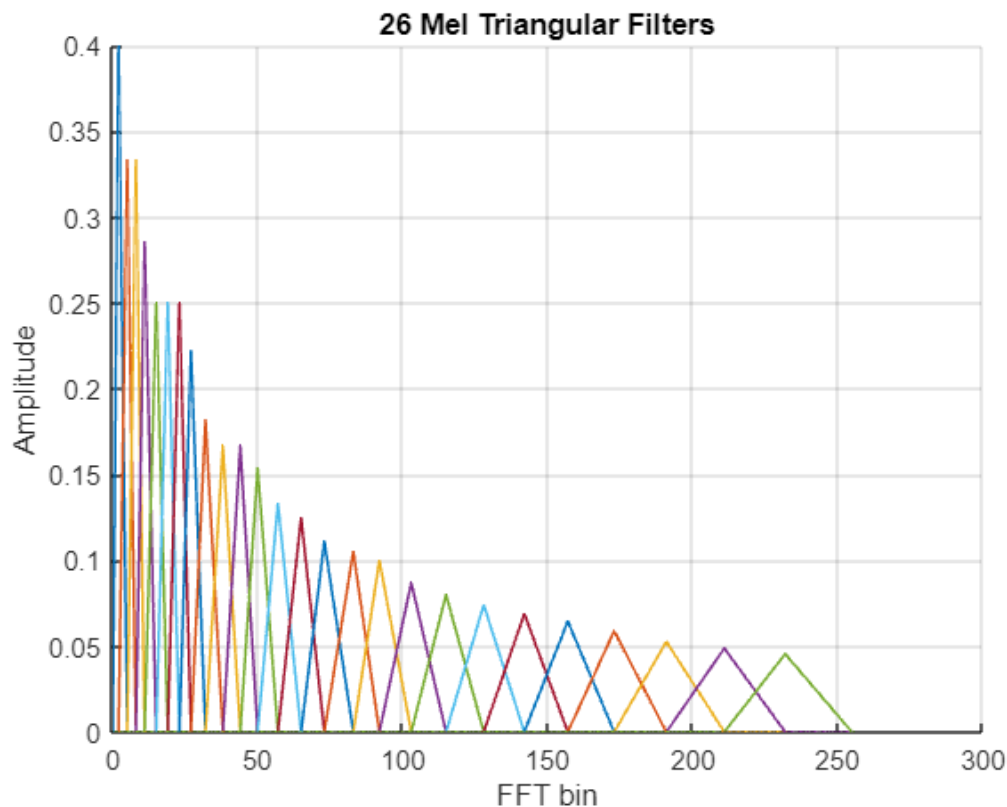
```

[num_filters, num_bins] = size(Melbank_filter);

for m = 1:num_filters
    plot(0:num_bins-1, Melbank_filter(m,:), 'LineWidth', 1);
end

xlabel('FFT bin');
ylabel('Amplitude');
title('26 Mel Triangular Filters');
grid on;
hold off;

```



```
mel_energies = apply_mel_filterbank(power_spectrum, fs, nfft, num_filters)
```

```

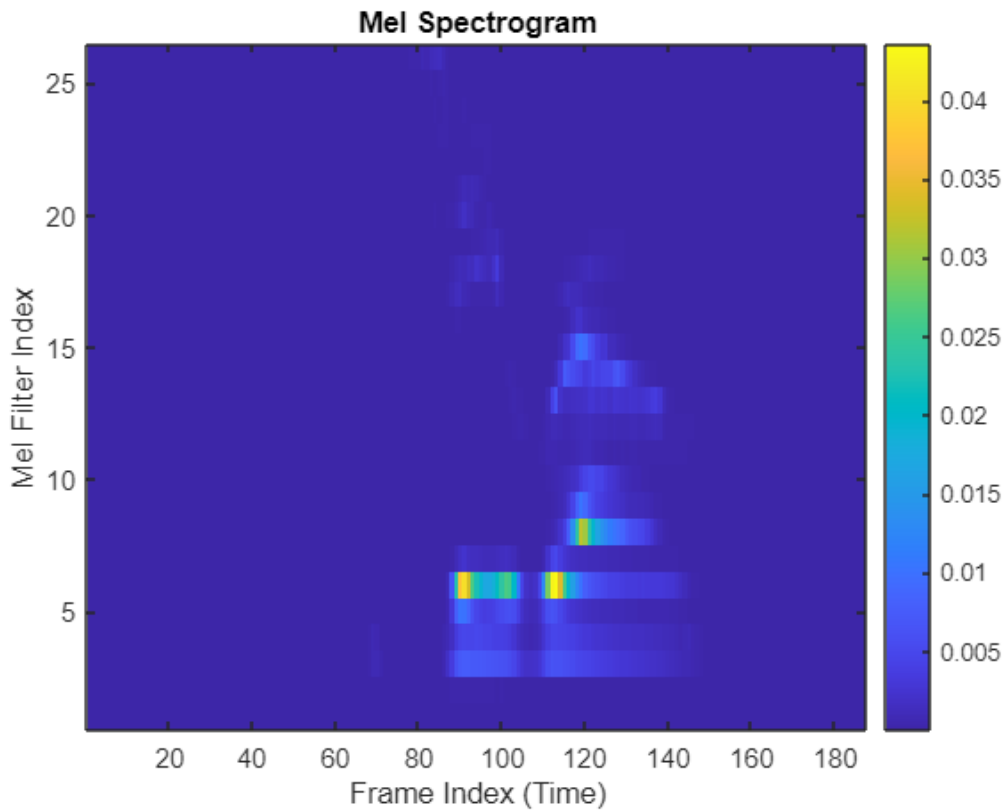
mel_energies = 26x187
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000 ...
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
    ...

```

Mel Energy

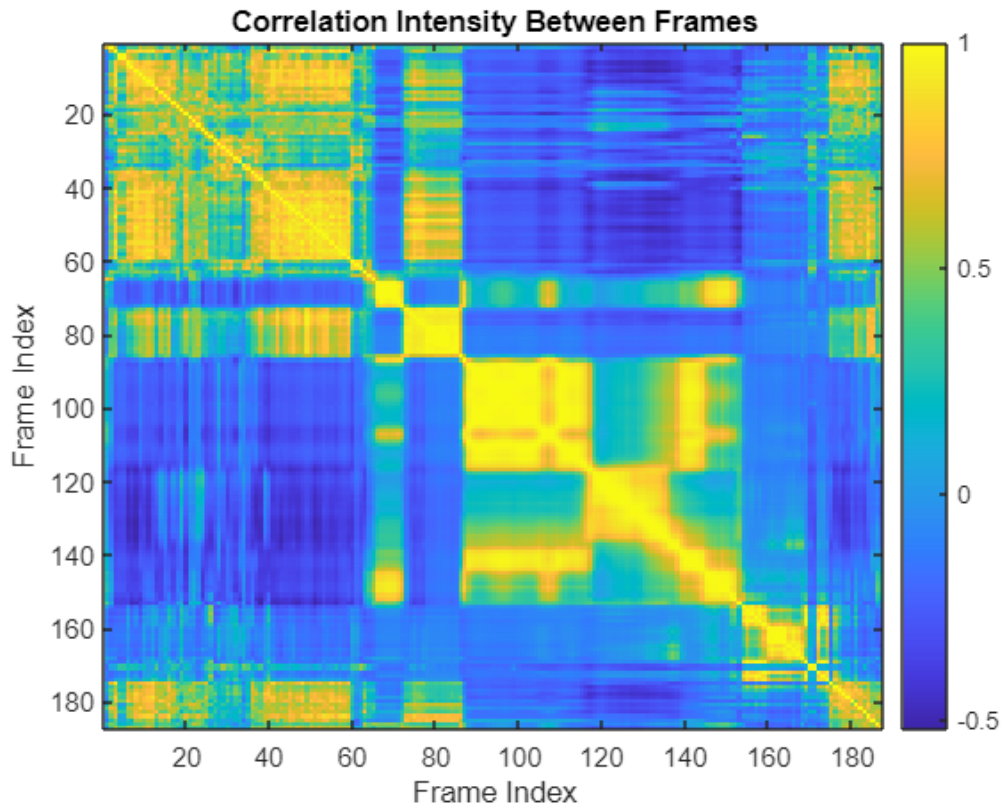
```
% Plot as an image
imagesc(mel_energies);

% Adjust axes
axis xy; % put low frequencies at bottom
xlabel('Frame Index (Time)');
ylabel('Mel Filter Index');
title('Mel Spectrogram');
colorbar;
```



Correlation across Frames

```
R = corrcoef(mel_energies); % correlation across frames
imagesc(R);
colorbar;
xlabel('Frame Index');
ylabel('Frame Index');
title('Correlation Intensity Between Frames');
```



Apply Log

```
log_mel = apply_log(mel_energies)
```

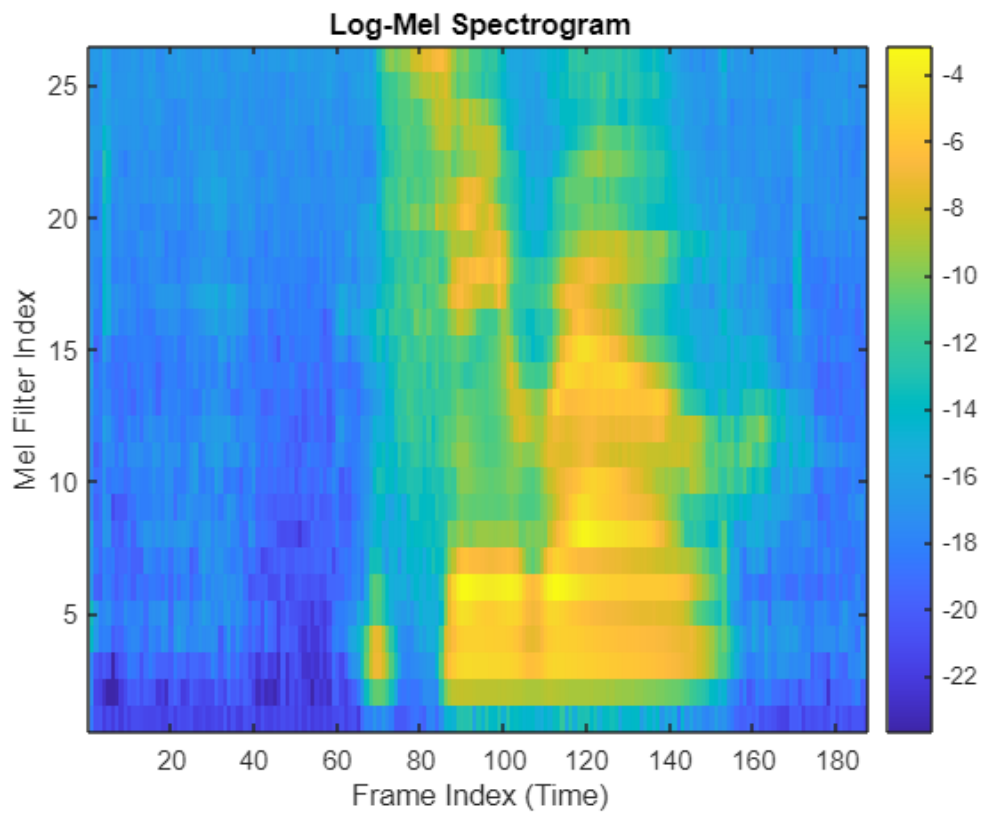
```
log_mel = 26x187
-18.5033 -20.7790 -20.5791 -21.9428 -21.1276 -22.0190 -20.8916 -22.1367 ...
-17.4673 -21.3604 -21.6382 -22.7959 -23.6812 -23.6670 -23.4893 -21.9964
-17.7948 -20.2586 -20.9840 -20.1582 -20.8246 -23.6149 -21.3135 -20.4045
-13.9581 -16.5849 -18.3196 -18.0312 -18.0820 -20.1150 -19.4175 -19.6294
-13.4247 -16.3370 -18.0255 -17.6427 -17.1409 -18.5406 -18.8897 -18.8388
-16.6198 -19.7256 -19.9608 -17.8100 -18.1578 -19.0504 -19.9844 -19.2378
-16.3879 -18.4325 -19.3966 -17.9174 -18.5720 -18.3069 -19.7267 -19.4056
-15.8839 -19.1571 -19.1786 -16.6927 -16.9527 -19.8188 -19.2811 -19.9371
-17.4075 -19.8105 -19.0898 -16.7304 -17.4547 -21.1375 -20.1152 -20.2067
-17.5053 -19.0326 -18.8475 -16.9758 -18.7498 -18.2646 -18.1552 -18.4922
⋮
```

```
% Plot as an image
imagesc(log_mel);

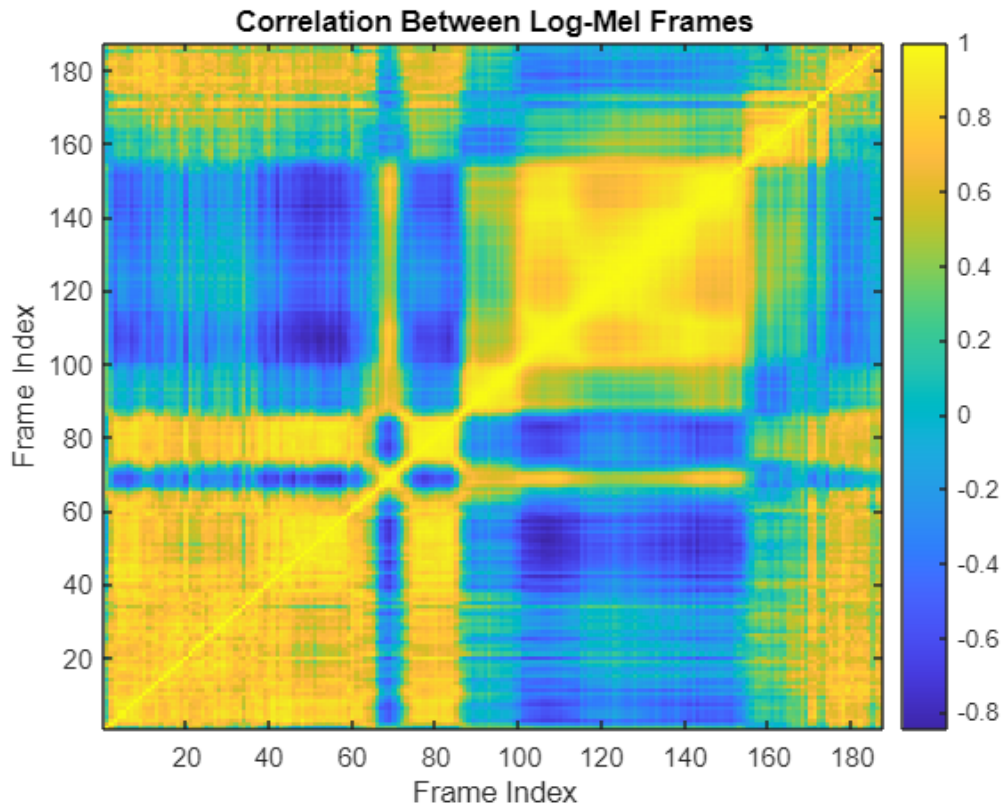
% Adjust axes
axis xy; % put low frequencies at bottom
xlabel('Frame Index (Time)');
ylabel('Mel Filter Index');
title('Log-Mel Spectrogram');
```



```
colorbar;
```



```
R_log = corrcoef(log_mel);  
  
imagesc(R_log);  
axis xy;  
colorbar;  
xlabel('Frame Index');  
ylabel('Frame Index');  
title('Correlation Between Log-Mel Frames');
```



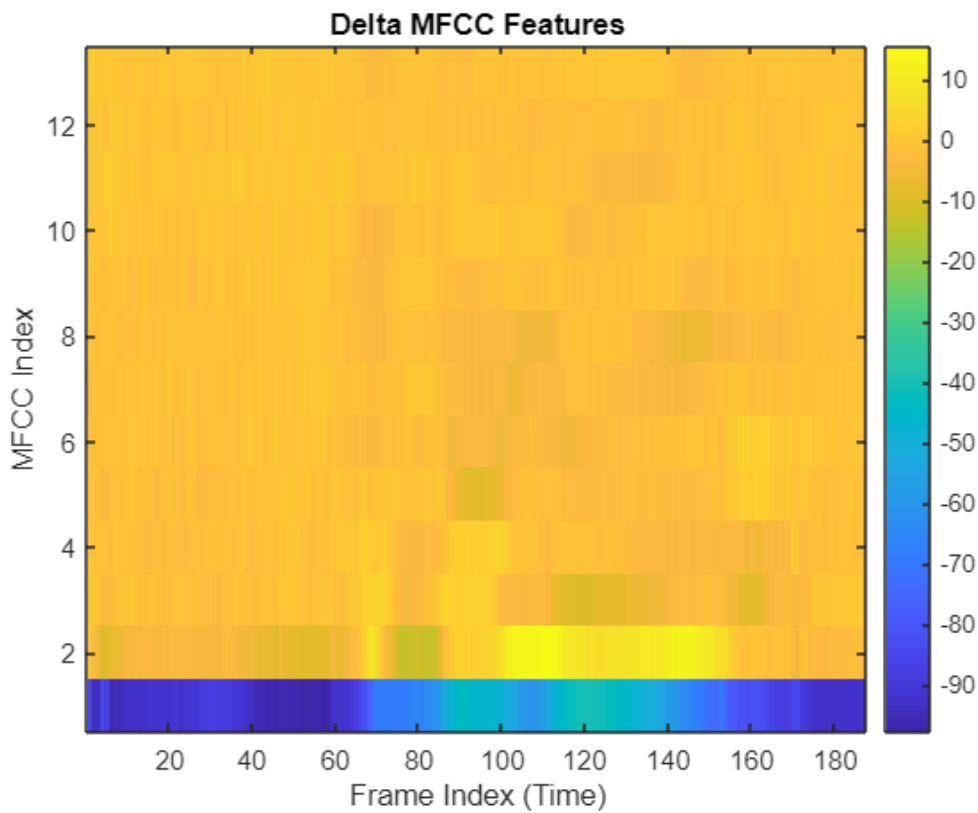
Apply Fourier Cosine transform

```
num_coeffs = 13;
mfcc = apply_dct(log_mel, num_coeffs)
```

```
mfcc = 13x187
-84.6655 -94.0774 -94.1807 -83.6803 -87.5592 -95.7119 -94.7470 -94.8710 ...
 0.2524 -4.0828 -5.6821 -10.5412 -8.8929 -7.3801 -7.1709 -6.6387
 0.4443 1.2478 0.1951 -3.2622 -2.2999 -2.2646 -1.3726 -0.7359
-0.7725 -0.9594 -1.4289 -0.8312 -0.9034 -2.5740 -1.1433 -1.0117
-1.0688 -1.4692 -0.9371 -4.5476 -4.0387 -1.9363 -1.1130 -1.6810
-2.5284 -2.2852 -0.5829 -0.9631 -1.5975 -1.2042 -1.0631 -0.9478
-2.9888 -1.8545 -1.2236 -0.9473 -1.2335 -1.6290 -0.7763 -0.7465
-0.5658 -1.8575 -0.9534 -0.5775 -0.4675 -0.7546 -0.2850 -1.6506
-0.8449 -1.1689 -0.4859 -1.7329 -1.3765 -0.6841 -1.5468 -1.1620
-0.7725 0.5384 0.0419 -0.4310 0.2572 2.1672 0.5291 0.0586
⋮
```

```
% Plot as an image
imagesc(mfcc)
% Adjust axes
axis xy; % put low frequencies at bottom
xlabel('Frame Index (Time)');
ylabel('MFCC Index');
title('Delta MFCC Features');
```

```
colorbar;
```

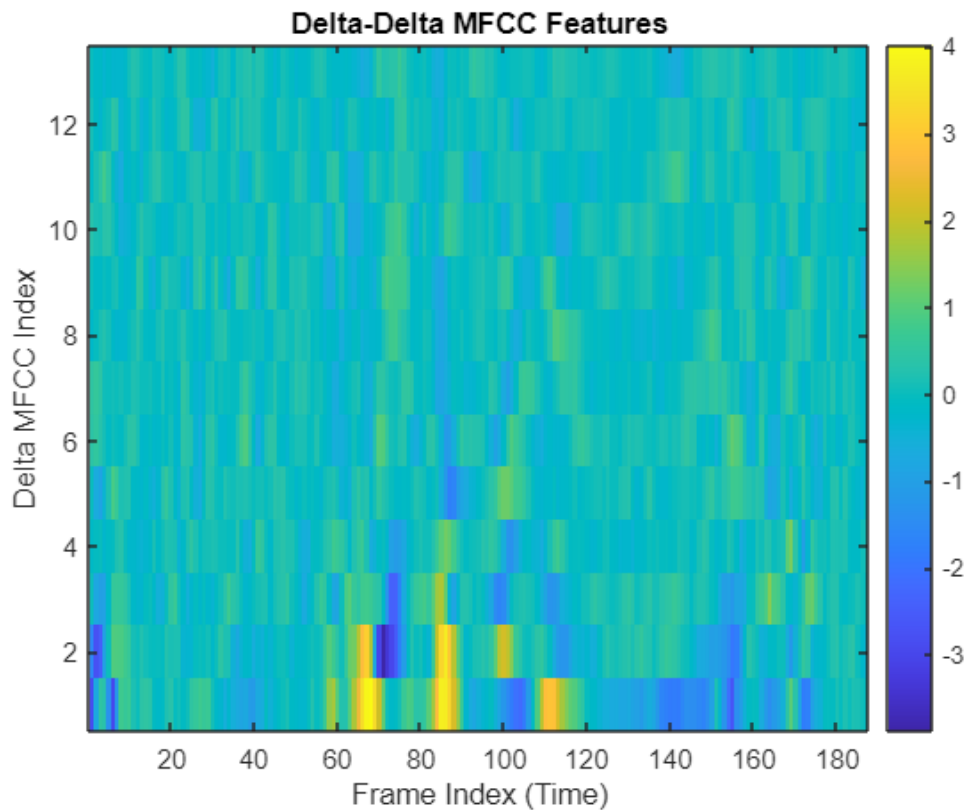


Apply Delta Fourier Cosine transform

```
delta_mfcc = compute_delta(mfcc)
```

```
delta_mfcc = 13x187
-2.8442 -0.7545 0.4610 0.3352 -1.3164 -2.9569 -1.1728 0.7417 ...
-1.6204 -2.7521 -2.4749 -0.9806 0.0183 0.9527 0.9096 0.9075
0.0305 -0.7662 -0.9998 -0.9520 -0.2138 0.5980 0.6154 0.5964
-0.1500 -0.0774 -0.0134 -0.2704 -0.1171 -0.0601 0.1995 0.2935
-0.0137 -0.6826 -0.9018 -0.4036 0.2259 0.8659 0.6486 0.1882
0.4134 0.5076 0.3184 0.1147 -0.1202 0.0565 -0.0423 -0.1090
0.4665 0.5848 0.4418 0.0441 0.0213 0.0859 0.0436 0.1188
-0.2067 -0.0411 0.1477 0.2692 0.1160 -0.1964 -0.3056 -0.2889
0.0394 -0.1417 -0.1627 0.0079 -0.1073 0.0972 0.1571 -0.1415
0.2940 0.1497 0.1090 0.3473 0.3573 0.1251 -0.1724 -0.4864
⋮
```

```
% Plot as an image
imagesc(delta_mfcc)
% Adjust axes
axis xy; % put low frequencies at bottom
xlabel('Frame Index (Time)');
ylabel('Delta MFCC Index');
title('Delta-Delta MFCC Features');
colorbar;
```

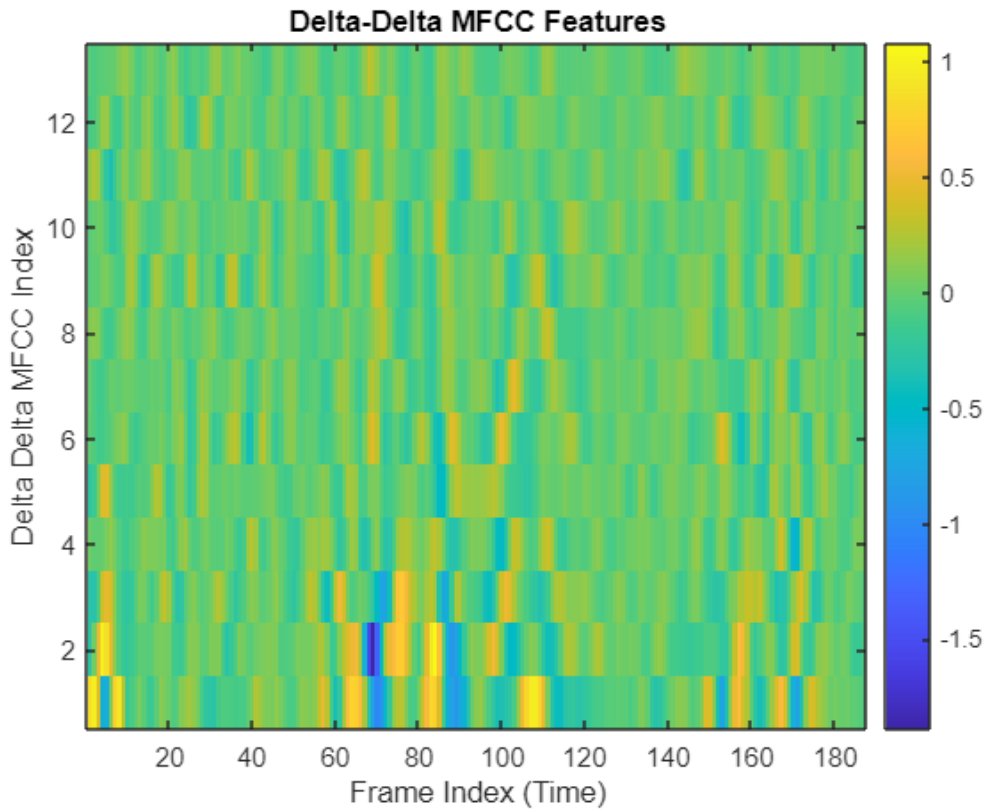


Apply Delta Delta Fourier Cosine transform

```
delta_delta_mfcc = compute_delta(delta_mfcc)
```

```
delta_delta_mfcc = 13x187
    0.8700    0.9664    0.4145   -0.6182   -0.6560    0.0957    0.7557    0.9085 ...
   -0.2841    0.0425    0.5049    0.9903    0.8702    0.4667    0.1418   -0.0836
   -0.2857   -0.2995   -0.0674    0.3515    0.4780    0.3926    0.1369   -0.0949
    0.0346   -0.0104   -0.0127   -0.0069    0.0636    0.1444    0.0586   -0.0555
   -0.2445   -0.1668    0.0758    0.4225    0.4370    0.1606   -0.0616   -0.1584
   -0.0096   -0.0692   -0.1460   -0.1341   -0.0780   -0.0370    0.0346    0.0818
    0.0069   -0.0869   -0.1431   -0.1418   -0.0755    0.0172   -0.0039   -0.0206
    0.0874    0.1306    0.0956   -0.0342   -0.1372   -0.1538   -0.0873    0.0372
   -0.0585   -0.0265   -0.0144    0.0533    0.0729   -0.0034   -0.0594   -0.1122
   -0.0514   -0.0078    0.0324    0.0199   -0.0785   -0.2197   -0.2182   -0.1227
    ...
    ...
    ...
```

```
% Plot as an image
imagesc(delta_delta_mfcc)
% Adjust axes
axis xy; % put low frequencies at bottom
xlabel('Frame Index (Time)');
ylabel('Delta Delta MFCC Index');
title('Delta-Delta MFCC Features');
colorbar;
```



Final Feature Matrix

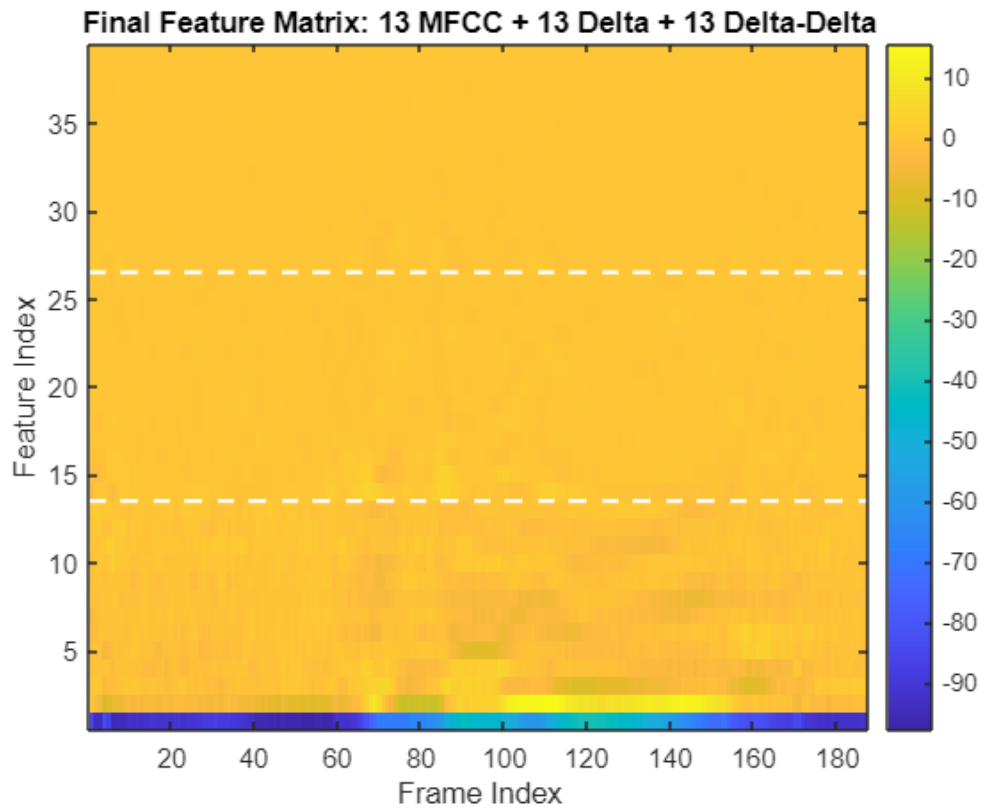
```
Final_feature_vector=[mfcc;delta_mfcc;delta_delta_mfcc]
```

```
Final_feature_vector = 39x187
-84.6655 -94.0774 -94.1807 -83.6803 -87.5592 -95.7119 -94.7470 -94.8710 ...
 0.2524 -4.0828 -5.6821 -10.5412 -8.8929 -7.3801 -7.1709 -6.6387
 0.4443 1.2478 0.1951 -3.2622 -2.2999 -2.2646 -1.3726 -0.7359
-0.7725 -0.9594 -1.4289 -0.8312 -0.9034 -2.5740 -1.1433 -1.0117
-1.0688 -1.4692 -0.9371 -4.5476 -4.0387 -1.9363 -1.1130 -1.6810
-2.5284 -2.2852 -0.5829 -0.9631 -1.5975 -1.2042 -1.0631 -0.9478
-2.9888 -1.8545 -1.2236 -0.9473 -1.2335 -1.6290 -0.7763 -0.7465
-0.5658 -1.8575 -0.9534 -0.5775 -0.4675 -0.7546 -0.2850 -1.6506
-0.8449 -1.1689 -0.4859 -1.7329 -1.3765 -0.6841 -1.5468 -1.1620
-0.7725 0.5384 0.0419 -0.4310 0.2572 2.1672 0.5291 0.0586
⋮
```

```
imagesc(Final_feature_vector);
axis xy;
xlabel('Frame Index');
ylabel('Feature Index');
title('Final Feature Matrix: 13 MFCC + 13 Delta + 13 Delta-Delta');
colorbar;

hold on;
plot([0.5 size(Final_feature_vector,2)+0.5], [13.5 13.5], 'w--', 'LineWidth', 1.5);
```

```
plot([0.5 size(Final_feature_vector,2)+0.5], [26.5 26.5], 'w--', 'LineWidth', 1.5);  
hold off;
```



```
matlab.internal.liveeditor.openAndConvert('Final_assemble.mlx',  
'Final_assemble.pdf');
```